

Fluid Properties (Ch. 1–3)

$$\rho = m/V, \gamma = \rho g, SG = \rho/\rho_w$$

$$\tau = \mu \frac{du}{dy} - \text{Newton's law of viscosity}$$

$$\nu = \mu/\rho - \text{kinematic viscosity}$$

$$h = \frac{4\sigma \cos \theta}{\gamma d} - \text{capillary rise, wetting if } \theta < 90^\circ$$

$$\Delta p = 2\sigma/R \text{ (droplet)}, 4\sigma/R \text{ (soap bubble)}$$

$$E_v = -V \frac{dp}{dV} = \rho \frac{dp}{d\rho} - \text{bulk modulus; } c = \sqrt{E_v/\rho}$$

$$\sigma_v = \frac{p - p_v}{\frac{1}{2}\rho V^2} - \text{cavitation number}$$

$$Re = \rho V L / \mu - \text{inertia/viscous; } Re_{cr} \approx 2300 \text{ (pipe)}, 5 \times 10^5 \text{ (plate)}$$

$$Fr = V/\sqrt{gL} - \text{inertia/gravity}$$

$$We = \rho V^2 L / \sigma - \text{inertia/surface tension}$$

$$Ma = V/c, \quad Eu = \Delta p / \rho V^2$$

$$\rho_w = 1000 \text{ kg/m}^3, \mu_w \approx 1.12 \times 10^{-3} \text{ Pa}\cdot\text{s}$$

$$\rho_{air} = 1.225 \text{ kg/m}^3, \mu_{air} \approx 1.79 \times 10^{-5} \text{ Pa}\cdot\text{s}$$

$$g = 9.81 \text{ m/s}^2, p_{atm} = 101.3 \text{ kPa}$$

Fluid Statics (Ch. 2)

$$\frac{dp}{dz} = -\gamma, \quad p = p_0 + \gamma h - \text{pressure vs. depth}$$

$$p_{abs} = p_{gauge} + p_{atm}$$

Manometer: $+\gamma h$ each descent, $-\gamma h$ each rise, tracing to the point of known pressure

$$F_R = \gamma h_c A - \text{resultant force, submerged plane surface}$$

$$y_{cp} = y_c + \frac{I_{xc}}{y_c A} - \text{center of pressure below centroid}$$

$$F_H = \text{force on the vertical projection of a curved surface}$$

$$F_V = \gamma V_{above} - \text{weight of real/virtual fluid above it}$$

$$F_R = \sqrt{F_H^2 + F_V^2} - \text{resultant on curved surface}$$

$$F_B = \gamma_{fluid} V_{displaced} - \text{buoyancy, acts at centroid of displaced volume}$$

$$\frac{\partial p}{\partial x} = -\rho a_x, \quad \frac{\partial p}{\partial z} = -\rho(g + a_z) - \text{linear accel.}$$

$$\frac{\partial p}{\partial r} = \rho r \omega^2 - \text{rigid-body rotation}$$

$$z_{surf} = \frac{\omega^2 r^2}{2g} + C - \text{paraboloid free surface}$$

Bernoulli & Euler (Ch. 4)

$$-\frac{\partial p}{\partial s} - \gamma \frac{\partial z}{\partial s} = \rho a_s - \text{Euler's eqn (streamline coord. } s)$$

$$\frac{p_1}{\gamma} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\gamma} + \frac{V_2^2}{2g} + z_2$$

steady, incompressible, inviscid, along a streamline, no pump/turbine between 1 and 2

$$p_0 = p + \frac{1}{2}\rho V^2 - \text{stagnation pressure}$$

$$V = \sqrt{2(p_0 - p)/\rho} - \text{from a Pitot-static tube}$$

$$-\frac{\partial p}{\partial n} - \gamma \frac{\partial z}{\partial n} = \frac{\rho V^2}{R} - \text{across curved streamlines (} n \rightarrow \text{center)}$$

Standard Bernoulli applications:

$$V = \sqrt{2gh} - \text{Torricelli, draining tank/orifice, } h = \text{head above orifice}$$

$$Q = A_2 \sqrt{\frac{2(p_1 - p_2)/\rho}{1 - (A_2/A_1)^2}} - \text{Venturi meter (ideal)}$$

$$Q_{actual} = C_d Q_{ideal} - \text{real orifice/venturi, } C_d < 1 \text{ from calibration}$$

Continuity (Ch. 5)

$$\sum \dot{m}_{in} = \sum \dot{m}_{out} - \text{steady flow, any CV}$$

$$A_1 V_1 = A_2 V_2 - \text{incompressible, single in/out}$$

$$\dot{m} = \rho A V_{avg}, \quad \dot{V} = \int V dA = V_{avg} A$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0 - \text{differential form}$$

$$\nabla \cdot \mathbf{V} = 0 - \text{incompressible flow}$$

$$\sum (\rho A V)_{in} = \sum (\rho A V)_{out} - \text{junctions, branching pipes}$$

Momentum (Ch. 6)

$$\sum \mathbf{F} = \sum \dot{m}_{out} \mathbf{V}_{out} - \sum \dot{m}_{in} \mathbf{V}_{in} - \text{steady CV, linear momentum}$$

$$F_x + p_1 A_1 - p_2 A_2 \cos \theta = \dot{m}(V_2 \cos \theta - V_1) - \text{pipe bend/nozzle, apply componentwise}$$

$$F_x = \dot{m}(V_2 \cos \theta - V_1)$$

$$F_y = \dot{m} V_2 \sin \theta - \text{force on a stationary vane/blade, deflection angle } \theta$$

$$\rho \left(\frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V} \cdot \nabla) \mathbf{V} \right) = -\nabla p + \rho \mathbf{g} + \mu \nabla^2 \mathbf{V}$$

Navier–Stokes, incompressible Newtonian fluid

$$T = \dot{m}(r_2 V_{t2} - r_1 V_{t1}) - \text{angular momentum / Euler turbomachine eqn}$$

Quick Reference — most-used

$$p_1/\gamma + V_1^2/2g + z_1 = p_2/\gamma + V_2^2/2g + z_2 - \text{Bernoulli}$$

$$A_1 V_1 = A_2 V_2 - \text{continuity} \quad Re = \rho V D / \mu - \text{pipe } Re$$

$$\sum \mathbf{F} = \dot{m}(\mathbf{V}_{out} - \mathbf{V}_{in}) - \text{momentum}$$

$$F_R = \gamma h_c A - \text{hydrostatic force} \quad F_B = \gamma_f V_{disp} - \text{buoyancy}$$

AER316 Fluid Mechanics — Final sheet, page 1 of 2. Ch. 1–6 (identical to the midterm sheet).

Energy Equation (Ch. 7)

$$\frac{p_1}{\gamma} + \frac{V_1^2}{2g} + z_1 + h_p = \frac{p_2}{\gamma} + \frac{V_2^2}{2g} + z_2 + h_t + h_L$$

h_p = pump head added, h_t = turbine head extracted, h_L = total head loss

$$P_{p,in} = \frac{\gamma \dot{V} h_p}{\eta_p}, \quad P_{t,out} = \eta_t \gamma \dot{V} h_t$$

$$HGL = p/\gamma + z, \quad EGL = HGL + V^2/2g$$

EGL always slopes down ($-h_L$); jumps $+h_p$ at a pump, drops $-h_t$ at a turbine

$\alpha = \int V^3 dA / (V_{avg}^3 A)$ – KE correction factor; ≈ 2.0 laminar, ≈ 1.05 turbulent

Dimensional Analysis (Ch. 8)

$n_\pi = n - m$ – Buckingham II theorem (n vars, m primary dims)

Method: choose m repeating variables spanning all dimensions, combine each remaining variable with them into a π group.

$$C_D = \frac{F_D}{\frac{1}{2}\rho V^2 A}, \quad C_L = \frac{F_L}{\frac{1}{2}\rho V^2 A} - \text{drag / lift coefficients}$$

$$St = fL/V - \text{Strouhal (unsteadiness)}, \quad C_p = \frac{p - p_\infty}{\frac{1}{2}\rho V^2}$$

Similarity: geometric (shape) / kinematic (velocity field) / dynamic (all π 's match)

$$\left(\frac{VL}{\nu}\right)_m = \left(\frac{VL}{\nu}\right)_p - \text{Reynolds scaling}$$

$$\left(\frac{V}{\sqrt{gL}}\right)_m = \left(\frac{V}{\sqrt{gL}}\right)_p - \text{Froude scaling}$$

Re and Fr scaling conflict – match whichever force dominates: Fr for free-surface/ship models, Re for fully-submerged flow

Surface Resistance — Boundary Layers (Ch. 9)

δ : $u(\delta) = 0.99U$ – 99% boundary-layer thickness

$$\delta^* = \int_0^\infty \left(1 - \frac{u}{U}\right) dy - \text{displacement thickness}$$

$$\theta = \int_0^\infty \frac{u}{U} \left(1 - \frac{u}{U}\right) dy - \text{momentum thickness}$$

Blasius (laminar flat plate):

$$\delta/x = 5.0/\sqrt{Re_x}$$

$$\delta^*/x = 1.721/\sqrt{Re_x}$$

$$\theta/x = C_{f,x} = 0.664/\sqrt{Re_x}$$

$C_{f,avg} = 1.328/\sqrt{Re_L}$ – average skin friction, laminar

Turbulent (empirical, flat plate, $5 \times 10^5 < Re_x < 10^7$):

$$\delta/x \approx 0.37/Re_x^{0.2}$$

$$C_{f,x} \approx 0.0592/Re_x^{0.2}$$

$$C_{f,avg} \approx 0.074/Re_L^{0.2}$$

$$F_D = C_{f,avg} \left(\frac{1}{2}\rho U^2\right) A - \text{skin-friction drag}$$

Separation: adverse gradient ($dp/dx > 0$) $\rightarrow \tau_w \rightarrow 0 \rightarrow$ separation.
Favorable gradient ($dp/dx < 0$) delays it.

Flow in Conduits (Ch. 10)

$Re = \rho V D / \mu = V D / \nu$ – pipe Reynolds number; laminar < 2300 ,

turbulent > 4000

Hagen–Poiseuille (laminar):

$$\dot{V} = \frac{\pi \Delta p D^4}{128 \mu L}, \quad V_{avg} = \frac{\Delta p D^2}{32 \mu L}, \quad f = 64/Re$$

$$h_f = f \frac{L}{D} \frac{V^2}{2g} - \text{Darcy–Weisbach (major loss, laminar or turbulent)}$$

Colebrook (implicit, turbulent):

$$\frac{1}{\sqrt{f}} = -2 \log_{10} \left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{Re \sqrt{f}} \right)$$

Swamee–Jain (explicit, $\pm 1\%$ of Colebrook):

$$f = \frac{0.25}{\left[\log_{10} \left(\frac{\varepsilon/D}{3.7} + \frac{5.74}{Re^{0.9}} \right) \right]^2}$$

Read f off the Moody chart (appendix) from Re & ε/D when a quick value will do.

$$h_L = K \frac{V^2}{2g} - \text{minor loss, } K = \text{fitting/valve loss coefficient}$$

$L_e/D = K/f$ – equivalent-length method

Typical K : entrance sharp 0.5, rounded 0.04; exit 1.0; elbow 90° std. 0.9, long-rad. 0.6; gate valve (open) 0.2; globe valve (open) 10

Pipe systems: series – same $\dot{V}$, $\sum h_L$ adds; parallel – same h_L , $\sum \dot{V}$ adds

Quick Reference — most-used, whole course

$$p_1/\gamma + V_1^2/2g + z_1 = p_2/\gamma + V_2^2/2g + z_2 - \text{Bernoulli}$$

$$A_1 V_1 = A_2 V_2 - \text{continuity}$$

$$Re = \rho V D / \mu \text{ (pipe), } \rho V L / \mu \text{ (general)}$$

$$\sum \mathbf{F} = \dot{m}(\mathbf{V}_{out} - \mathbf{V}_{in}) - \text{momentum}$$

$$h_f = f(L/D)(V^2/2g), \quad h_{L,minor} = K(V^2/2g)$$

$$F_R = \gamma h_c A - \text{hydrostatic force; } F_B = \gamma_f V_{disp} - \text{buoyancy}$$

$$C_D, C_L = F / (\frac{1}{2}\rho V^2 A) - \text{drag/lift coefficients}$$

$n_\pi = n - m$ – Buckingham II

Solving a Problem — which equation?

Steady, no friction/pump/turbine, along a streamline $\rightarrow$ Bernoulli.

Friction, a pump, or a turbine between two sections $\rightarrow$ full energy equation.

Need a force on a fitting, vane, or support $\rightarrow$ momentum equation.

Comparing a scale model to a prototype $\rightarrow$ dimensional analysis / similitude.

Flow over a flat surface, need drag or where flow separates $\rightarrow$ boundary-layer relations.

Flow through a pipe, need pressure drop or pump sizing $\rightarrow$ Darcy–Weisbach + Moody/Colebrook + minor losses.

AER316 Fluid Mechanics — Final sheet, page 2 of 2. Condensed from the 58-formula verified reference in the Formula_Ref app — keeps what you're most likely to reach for mid-problem.